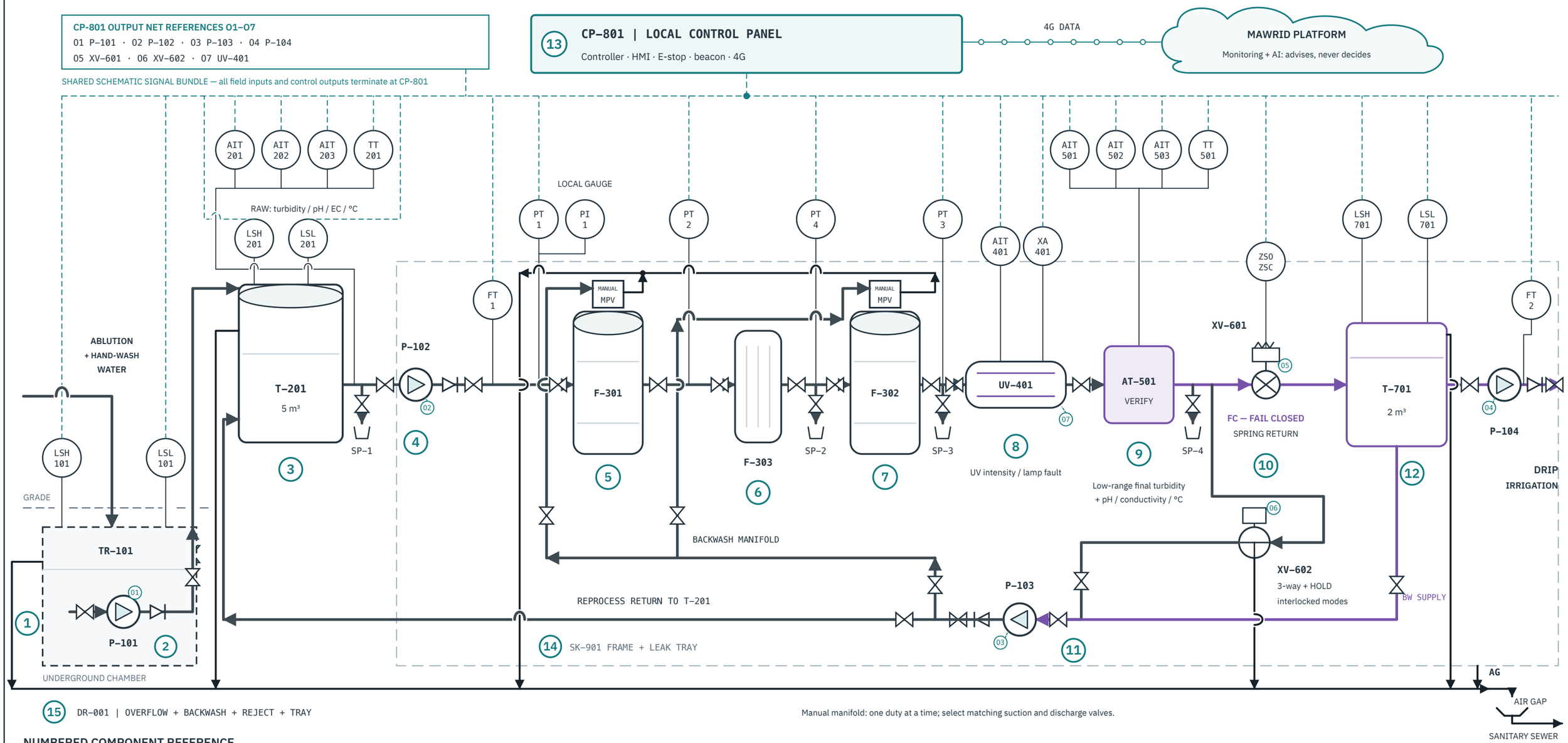



15 DR–001 | OVERFLOW + BACKWASH + REJECT + TRAY

Manual manifold: one duty at a time; select matching suction and discharge valves.

## NUMBERED COMPONENT REFERENCE

| | | |
|---|---|---|
| 1 | TR–101 | Collection chamber & trap |
| 2 | P–101 | Submersible lift pump |
| 3 | T–201 | Equalization tank |
| 4 | P–102 | Feed pump |
| 5 | F–301 | Multimedia sand filter |
| 6 | F–303 | 5 µm PP cartridge filter |
| 7 | F–302 | Carbon / palm Biochar filter |
| 8 | UV–401 | UV disinfection reactor |
| 9 | AT–501 | Verification cell |
| 10 | XV–601 / XV–602 | Irrigation + diversion valves |
| 11 | P–103 | Backwash / reprocess pump |
| 12 | T–701 / P–104 | Treated-water tank + irrigation pump |
| 13 | CP–801 | Control panel |
| 14 | SK–901 | Frame + leak tray |
| 15 | DR–001 | Drain lines |

RAW / PROCESS — TREATED — DRAIN — SIGNAL — ○ field ⊖ panel ∩ no connection

## LOCAL PROTECTION LOGIC

| CONDITION | RESPONSE |
|---|---|
| Power loss | XV-601 spring-closes; all pumps stop. |
| Final turbidity above limit | XV-602 → reprocess; after 3 attempts: HOLD + alarm. |
| UV <70% or lamp fault | Immediate HOLD + alarm + maintenance request. |
| Critical sensor lost / frozen / low confidence | Immediate HOLD + calibration request. |
| High filter differential pressure | Backwash; if persistent, issue a service alert. |
| T-201 low level | Stop P-102 to prevent dry running. |
| T-701 full | Pause the treatment train. |
| XV-601 closure not confirmed | Critical alarm + stop all pumps. |

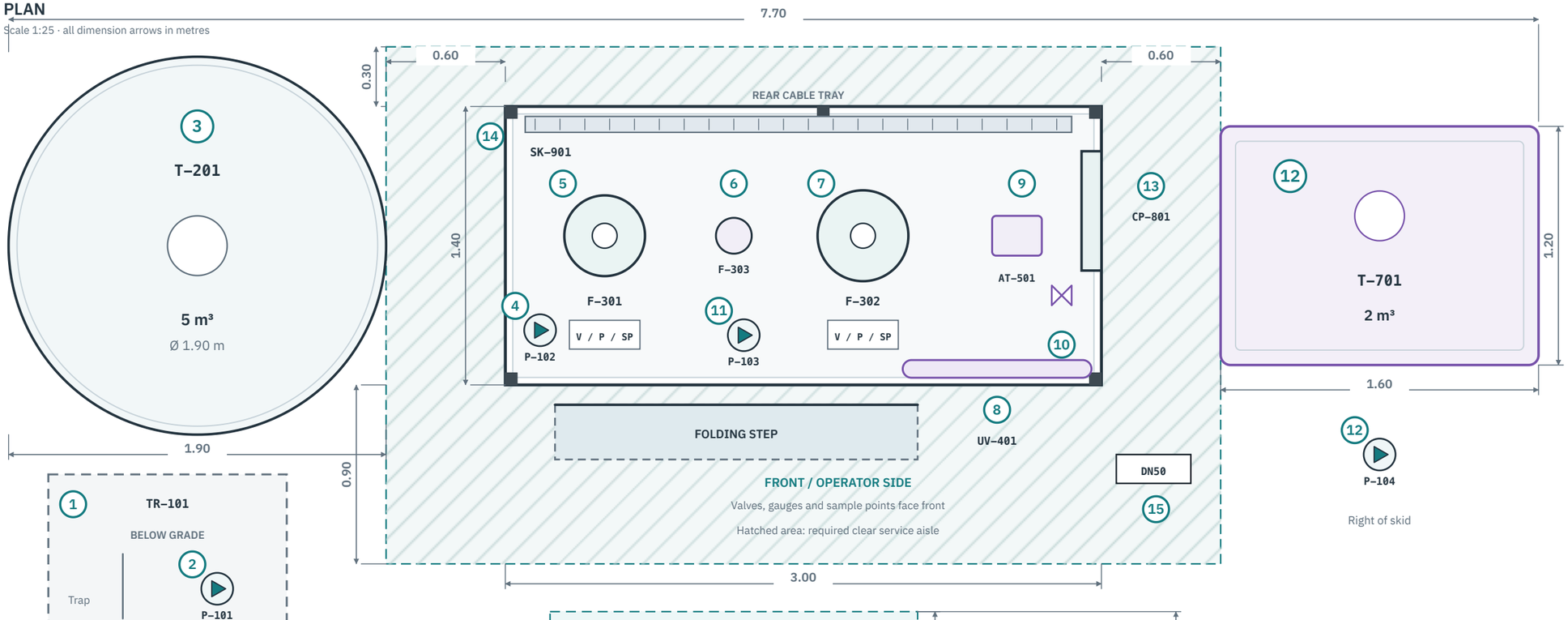
**Default / any fault: XV-601 CLOSED. HOLD: XV-602 shut; pumps stopped.**
**Fixed rules in the local controller authorize irrigation, even offline.**
**AI may only make the decision more conservative.**

**PROJECT: MAWRID | SIMPLIFIED P&ID**
1/3 | REV A | 2026–10–15 | NTS | COMPETITION CONCEPT

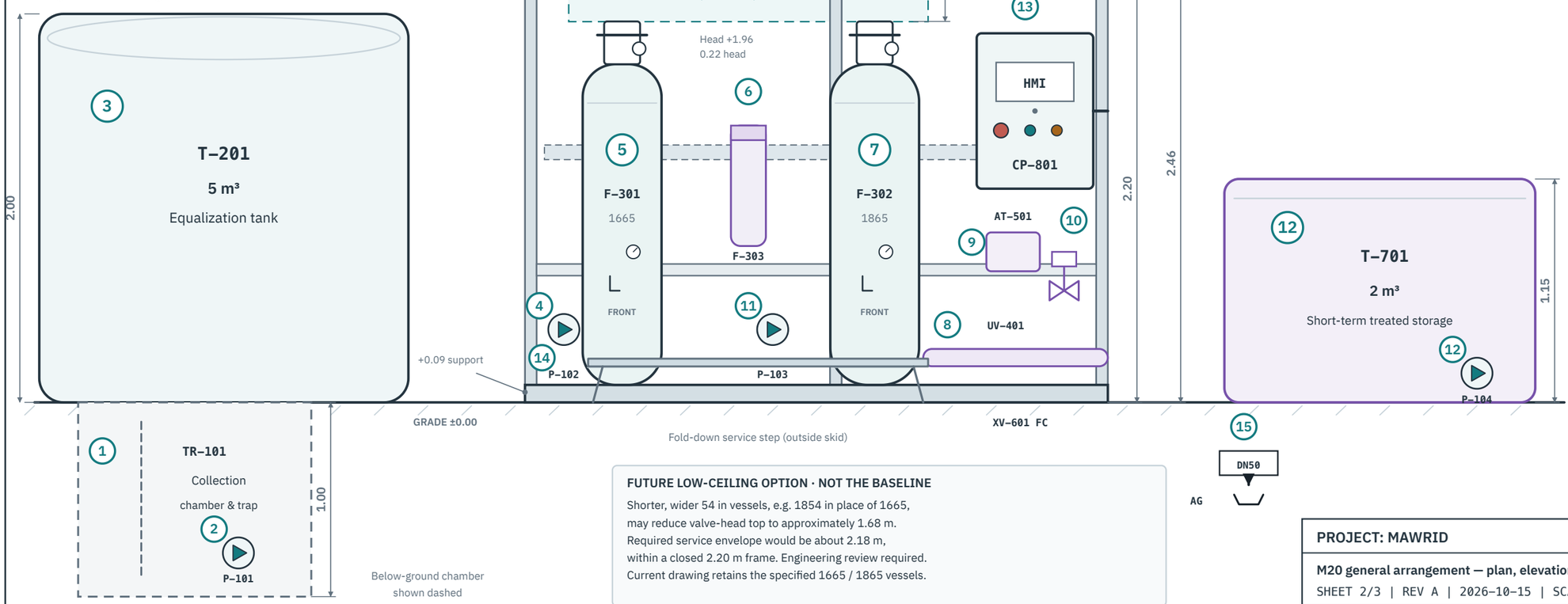
Competition concept — not approved for construction. Dimensions approximate, based on common commercial sizes.

## PLAN

Scale 1:25 · all dimension arrows in metres




## FRONT ELEVATION




### FUTURE LOW-CEILING OPTION · NOT THE BASELINE

Shorter, wider 54 in vessels, e.g. 1854 in place of 1665, may reduce valve-head top to approximately 1.68 m. Required service envelope would be about 2.18 m, within a closed 2.20 m frame. Engineering review required. Current drawing retains the specified 1665 / 1865 vessels.

## COMPONENT KEY

1. **TR-101** — Collection chamber & trap
2. **P-101** — Submersible lift pump
3. **T-201** — Equalization tank
4. **P-102** — Feed pump
5. **F-301** — Multimedia sand filter
6. **F-303** — 5 µm PP cartridge filter
7. **F-302** — Carbon / palm Biochar filter
8. **UV-401** — UV disinfection reactor
9. **AT-501** — Verification cell
10. **XV-601 / XV-602** — Irrigation + diversion valves
11. **P-103** — Backwash / reprocess pump
12. **T-701 / P-104** — Treated-water tank + irrigation pump
13. **CP-801** — Control panel
14. **SK-901** — Frame + leak tray
15. **DR-001** — Drain lines

## EQUIPMENT DIMENSIONS · m

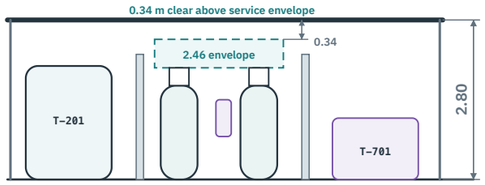
SK-901 3.00 × 1.40; 0.06 posts; 2.20 high
Bearing level +0.09 above grade
F-301 Ø0.406 × 1.651 shell (1665)
F-302 Ø0.457 × 1.651 shell (1865)
Valve heads 0.22 high; top +1.96
F-303 Ø0.180 × 0.620
UV-401 Ø0.089 × 0.950 long
T-201 Ø1.90 × 2.00 (5 m³)
T-701 1.60 × 1.20 × 1.15 (2 m³)
TR-101 1.20 × 0.80 × 1.00 deep
CP-801 0.60 × 0.80; centre +1.50
Process piping DN32; drain piping DN50

## DOCUMENTED CONSTRAINTS

01 Service aisles are outside the skid.
02 Heads at 1.96 m; use the folding step.
03 Open above filters: no overhead beams.
04 Corner/rear posts; CP on side/rear post.
05 Cable tray stays behind the filters.
06 Minimum clear site height: 2.50 m.
Indoors: verify equipment-room ceiling.
Outdoors: canopy >2.50 m or removable.
07 Final dimensions require engineering.

## OUTDOOR SITE · NTS

Mecca sun: shade polyethylene tanks, pipes and electronics




Shade is above the 2.50 m minimum clear site height.

| PROJECT: MAWRID | STATUS: COMPETITION CONCEPT |
|---|---|
| **M20 general arrangement — plan, elevation and site** | |
| SHEET 2/3 \| REV A \| 2026-10-15 \| SCALE 1:25 | |

Print A3 at 100% • do not scale a fitted-page print

Competition concept — not approved for construction. Dimensions approximate, based on common commercial sizes.